

Comparison of Low Level Processing for Binary Classification of Retinal Diseases

1st Raisa Tafannum

Department of CSE

Independent University, Bangladesh

Dhaka, Bangladesh

i.2211425@iub.edu.bd

2nd Jugumaya Saha Disha

Department of CSE

Independent University, Bangladesh

Dhaka, Bangladesh

ii.2320434@iub.edu.bd

Abstract—This paper compares image pre-processing methods of binary retinal disease classification based on deep learning. Diseases like diabetic retinopathy and macular degeneration should be detected early so that the vision loss can be prevented. Although networks such as ResNet50 are effective, their performance is sensitive to the quality of the input images. This research paper compares Sobel, 8-neighbourhood Laplacian and logarithmic transformations to improve the retinal images. With the help of the RFMiD data, the binary classification (disease vs normal) is simplified. Findings indicate that proper preprocessing enhances accuracy, recall and generalization, emphasizing its role in the development of effective and efficient retinal disease detecting systems.

I. INTRODUCTION

A. Effect on Life / Society

Permanent blindness and visual impairment are currently a global health issue of concern mainly caused by Retinal Diseases. A wide range of retinal diseases, such as diabetic retinopathy, glaucoma, and age-related macular degeneration affects millions of people every year. The rise of these diseases are due to aging trends, unhealthy lifestyles, and the increased number of chronic diseases. Vision impairment reduces quality of life and makes everyday tasks more difficult for a person. At worst, it may result in total blindness, triggering emotional, social and economic costs to the patients and their families. Most of the retinal diseases are either treatable or preventable when diagnosed at an early age. Hence, there is an immediate requirement of effective, precise and scalable screening systems that could help medical practitioners to diagnose retinal illnesses and alleviate the general strain on health care systems.

B. Traditional Method their drawbacks

Retinal diseases are conventionally diagnosed by ophthalmologists by the manual analysis of retinal fundus images, by means of the ophthalmoscopy technique and fundus photography. The most popular and common method is medical experts scrutinize the images keenly to identify abnormalities, including lesions, haemorrhages, exudates, and structural changes in the retina. Although this method is clinically valid, it poses a number of limitations intrinsically. To begin with, it is a time consuming process that needs highly trained experts making it inappropriate when handling large scale screening programs. Secondly, because experts differ in experience and

interpretation, diagnosis can be different between experts, thus creating inter-observer variability. Also, in most places especially the rural and underdeveloped ones, there is a lack of qualified ophthalmologists hence getting timely diagnosis is not easy. With the increasing population of patients, manual screening becomes more ineffective and inefficient thereby leading to late and improper diagnosis and development of retinal diseases. To address these difficulties and automated solutions is necessary to help in quicker and more reliable diagnosis in modern society

C. Significant Limitations and Objective of the research

Automated systems have been developed based on machine and deep learning methods, although there are still a number of challenges in existing methods. Most methods are based on multi-class classification or detecting a particular disease, complicating the models and raising computational demands. Other challenges exist like the imbalance of classes, overfitting, and ineffective generalisation of the systems with different datasets reduce the effectiveness of these systems. In practice, rather than identifying a variety of disease classes it is often necessary to identify a retinal disease (present or absent) for initial and quick diagnosis. The binary classification will simply do this identification, simplifying the problem greatly and decreasing the computational complexity and model performance. Hence, the main goal of this study is to create a binary classification system to identify the retinal disease (disease vs normal) with the help of the RFMiD dataset. The suggested method is expected to achieve a better detection accuracy, better generalisation, low computational cost and develop a scalable and dependable system that would be able to provide early diagnosis as well as aid in the decision making processes of clinicians.

II. LITERATURE REVIEW

A. Description of relevant existing works

- **Automated Detection of Diabetic Retinopathy Using Custom Convolutional Neural Networks** The paper mainly concentrates on a custom CNN network to detect retinal disease (mostly DR) with fundus images. The pre-processing step is very low in nature and most

often standard operations like image resizing, normalization (scaling of pixels) and basic augmentation is performed. No sophisticated filtering methods (such as edge-enhancement or contrast-specific transforms) are highlighted, limiting enhancement of features at the input level. Training the model is realized with a relatively shallow custom CNN, which is trained on a Kaggle dataset, without pre-trained weights. The design consists of pooling, convolutional and fully connected layers. It can be easily trained, with binary classification and cross-entropy loss. The model has high accuracy (95percent), which is attributed to the small dataset size, but probably due to bias in the dataset. The weaknesses of the paper are mainly its absence of solid pre-processing and inability to generalize. The model is not sensitive to complex retinal variations given the limited nature of the dataset and its binary nature. Also, lack of methods to improve the features (such as vessel highlighting or contrast normalization) implies that the CNN has to learn all the information on the raw image, which decreases its efficiency. The model is also not adaptable to multi-disease conditions and to clinical variability in real-life conditions [1].

- **MGSCNN: Multi-Gradient Swarm Optimized CNN for Retinal Disease Classification** In this paper, a hybrid model consisting of CNN and Glowworm Swarm Optimization (GSO) is proposed to classify multi-retinal diseases. The pre-processing phase involves simple normalization and dataset-specific processing, and is not very heavily dependent on traditional image processing filters. Rather, focus has been in optimizing feature selection and not improving input images. The training method is more sophisticated than the typical CNNs. The CNN is a feature extractor, whereas the GSO algorithm is a feature parameter optimizer (weights or feature subsets). This hybrid training enhances classification accuracy (95) on the RFMiD dataset especially in the multi-class problems. Nevertheless, optimization process is much more complex in terms of computational complexity and training time. The most important missing element is efficiency and scalability. Although it performs well, the model is computationally intensive and it cannot be easily implemented into real-time systems. Also, the absence of a good pre-processing implies that the model relies much on optimization and not on input quality. The method might not generalize to other datasets because the method uses parameter tuning. This approach is more than complex where simpler tasks are involved such as binary classification. [2].
- **Deep Learning-Based Multi-Layer CNN for Retinal Disease Classification Using RFMiD Dataset** The paper will discuss deep CNN models (12-layer and 20-layer models) when classifying retinal diseases with the help of the RFMiD dataset. The pre-processing methods are simple and they are resizing, normalization and restricted

augmentation. There are no sophisticated filters (e.g. edge-detecting or intensity transforms) and thus the fine retinal features cannot be enhanced. The training plan is on enhancing the training depth to enhance feature extraction. The 20-layer CNN represents more complicated patterns as opposed to the 12-layer model. Nevertheless, more complex networks add overfitting particularly when there is a small amount of data. Optimized models exhibit a higher accuracy of about +89.5percent which is lower than the best performance attained. Overfitting and lack of generalization is the major gap. Lack of good pre-processing implies that the model can barely recognize minuscule structures such as micro-lesions. Moreover, more complex architectures are added but not guaranteed to yield better performance. The model is not efficient as well and needs to be tuned. In this paper, it is emphasized that there is no need just to increase depth, which requires more accurate input representation (pre-processing) and regularization. [4].

- **Retinal Disease Classification Using CNN and Transfer Learning (VGG16)** In this paper, a baseline CNN is compared with a transfer learning model (VGG16) in the classification of retinal diseases. The pre-processing phase contains resizing, normalization and potentially augmentation, but once again contains no sophisticated processing methods such as contrast control or edge detection. It is evident that the training method can be improved with the help of transfer learning. CNN model provides a high accuracy of about 83percent and VGG16 provides an even better accuracy of about 90.8percent due to pretrained features. Training can be done by fine-tuning of higher layers but freezing of lower layers. This minimizes the time of training and enhances features extraction. The loophole is on the disproportion of classes and insensitivity to diseases. Nevertheless, the model has low recall (0.65) of diseased cases, which is a missed significant positive samples. The model also continues to be based on simple pre-processing that restricts the ability to detect minute retinal features. It is also binary based, and is not scalable to multi-disease cases. Cost in terms of computation is moderate yet high as compared to basic CNN. [4].
- **Retinal Disease Detection Using EfficientNet and Hybrid Deep Learning Models** The advanced architectures such as EfficientNet, hybrid models (DenseNet + MobileNet) related to the retinal disease classification are discussed in this paper. The pre-processing step consists of standard normalization and augmentation, but does not heavily incorporate classical image enhancement algorithms. The training plan employs the transfer learning based on the state-of-the-art architectures. EfficientNet is almost 93-94 percent accurate and hybrid models are almost 92.9 percent. These are trained using fine tuning and have an advantage in terms of feature extraction in that they have deeper and more optimized architectures.

But the difference is in computational complexity and difficulty in deployment. These models are computationally intensive and thus cannot be used in low resource or real time applications. Also, the absence of the highly developed pre-processing implies that the models are fully using deep learning to extract features, which might not be the best according to medical images. Another trade-off is the accuracy and efficiency. [5].

B. Compare Methods (Focuses on Pre-processing)

TABLE I
COMPARISON OF PRE-PROCESSING METHODS

Pre-processing	Dataset	Accuracy	Efficiency	Limitation
Resize + Normalize + Augment	Kaggle	95%+	High	Basic enhancement
Noise Removal + CLAHE	RFMiD	95.09%	Medium	Complex pipeline
Resize + Normalize	RFMiD	89.59%	High	Weak preprocessing
Resize + Augment	Kaggle	90.8%	High	Imbalance issue
CLAHE + Enhancement	Fundus	93%	Low	High computational cost

C. Research Gaps of Existing Models

- **Lack of Standardization:** Various papers employ various preprocessing steps.
- **Insufficient Enhancement:** There are techniques that do not show small lesions.
- **High Computational Cost:** The methods (CLAHE, pipelines) are costly.
- **High Computational Cost:** The methods (CLAHE, pipelines) are costly.
- **Failing to address Class Imbalance:** Imbalance can hardly be completely addressed in pre-processing.
- **Overfitting Still Exists:** The poor generalization is caused by weak preprocessing.

To summarize, simple preprocessing techniques are less effective and computationally efficient than more sophisticated techniques. However, accuracy can be enhanced by such methods as CLAHE, which makes the methods more complex. Uniform, efficient and lightweight preprocessing pipelines are integral to achieve cost effective images of retinal disease .

III. DATASET

A. RFMiD Dataset

The Retinal Fundus Multi-Disease Image Dataset (RFMiD) is a broad-based open-source dataset with the aim of being useful in automated detection of retinal diseases through medical imaging. In contrast to previous datasets, which only contained a small number of known conditions (like diabetic retinopathy or glaucoma), RFMiD has a broad range of common and rare retinal pathologies, thus facilitating the creation of more clinically relevant and generalized artificial intelligence models. The data is made up of 3,200 retinal fundus images between 2009 and 2020, patients who receive regular or symptomatic eye exams. Multiple fundus cameras were used to obtain images, thereby introducing a variety of resolution and quality, which replicated the clinical situation in the real world. It is important to note that there are high-quality

and low-quality images in the dataset, which complicates the data and makes it more applicable in practice. RFMiD enables association of an image with multiple labels, thereby, reflecting the actual clinical situations. It is well-organized into subsets of the data (training 60 per cent, validation 20 per cent, testing 20 per cent) and contains csv file indicating and connecting each image with it's corresponding disease (class) label as shown on Fig. 1,.The annotations is checked by skilled ophthalmologists to maintain high fidelity. [6]

RFMiD includes a total of 46 classes, consisting of 1 normal class (NL) and 45 disease/pathology classes. These diseases include a wide range of retinal abnormalities such as Diabetic Retinopathy (DR), Age-related Macular Degeneration (ARMD), Media Haze (MH), Drusen (DN), Myopia (MYA), Branch Retinal Vein Occlusion (BRVO), Tessellation (TSLN), Epiretinal Membrane (ERM), Laser Scars (LS), Macular Scar (MS), Central Serous Retinopathy (CSR), Optic Disc Cupping (ODC), Central Retinal Vein Occlusion (CRVO), Tortuous Vessels (TV), Asteroid Hyalosis (AH), Optic Disc Pallor (ODP), Optic Disc Edema (ODE), Anterior Ischaemic Optic Neuropathy (AION), Retinitis (RS), Chorioretinitis (CRS), and many more rare conditions.

Although it has merits, RFMiD has such issues as imbalance of classes, multi-label complexity, and lack of demographic. It is, however, an invaluable resource towards the development of strong, scalable, and precise AI-based retinal disease detection technologies. [6].

B. RFMiD Dataset Details in Tabular Format

TABLE II
RFMiD DATASET DETAILS

Section	Details
Dataset Name	Retinal Fundus Multi-Disease Image Dataset (RFMiD)
Total Images	3200
No. of Subjects	Not mentioned
Dataset Type	Retinal Fundus Image Dataset
Task Type	Multi-label Classification
Data Split	Training: 60% (1920 images), Validation: 20% (640 images), Test: 20% (640 images)
Classes (All)	NL, DR, ARMD, MH, DN, MYA, BRVO, TSLN, ERM, LS, MS, CSR, ODC, CRVO, TV, AH, ODP, ODE, ST, AION, PT, RT, RS, CRS, EDN, RPEC, MHL, RP, CWS, CB, ODPM, PRH, MNF, HR, CRAO, TD, CME, PTCR, CF, VH, MCA, VS, BRAO, PLQ, HPED, CL
Subject Demography	Not mentioned
Age Range	Not mentioned
Annotation Process	Annotated by ophthalmologists and verified through expert consensus
Data Source Location	Eye clinic and research institute (India)
Image Acquisition	TOPCON 3D OCT-2000, Kowa VX-10, TOPCON TRC-NW300
Dataset Challenges	Highly imbalanced dataset
Use Case	AI-based retinal disease detection

ID	Diabetic_Risk	DR	ARMD	MH	DN	MYA	BRVO	TSLN	ERM	LS	MS	CSR
1	1	1	1	0	0	0	0	0	1	0	0	0
2	1	1	1	0	0	0	0	1	0	0	0	0
3	1	1	1	0	0	0	0	0	0	0	0	0
4	1	0	0	0	0	0	0	0	0	0	0	0
5	1	0	0	0	0	0	0	0	0	0	0	0
6	1	0	0	0	0	0	0	0	0	0	1	0
7	1	0	0	0	0	0	0	0	0	0	1	0
8	1	0	0	0	0	0	0	0	1	0	0	0
9	1	1	0	0	0	0	0	0	0	0	0	0
10	1	1	0	0	0	0	0	0	0	1	0	0

Fig. 1. Dataset Arrangement

C. Data Distribution

- **Normal images (NL):** 669
- **Diabetic Retinopathy (DR):** 632
- **Media Haze (MH):** 523
- **Optic Disc Cupping (ODC):** 445
- **Tessellation (TSLN):** 304
- **Drusen (DN):** 230
- **Age-related Macular Degeneration (ARMD):** 169
- **Myopia (MYA):** 167
- **Branch Retinal Vein Occlusion (BRVO):** 119
- **Other diseases:** 1–10 images

IV. METHODOLOGY

The proposed algorithm combines low-level image processing with a high-level deep learning structure that refines the feature representation and boosts the performance of retinal disease classification. Raw images in medical imaging especially retinal fundus analysis commonly face issues of poor contrast, non-uniform lighting, noise, and poor view of important parts of the body. Such constraints can greatly affect the process of machine learning models in learning discriminative features. Hence, a powerful preprocessing pipeline is needed so as to clean up the input data prior to model training.

The first step in the procedure entails classical preprocessing methods to improve structural and intensity-related properties of the pictures. These functions underline the main diagnostic characteristics, such as blood vessels, edges of lesions, and minor abnormalities, with contrast amplification and edges sharpness. This step enhances better visual interpretability but it also enables better feature extraction in the learning process.

The resulting processed images are then given to the ResNet50 neural network, a deep convolutional network that learns features in a hierarchy. ResNet50 takes advantage of residual connections and overcomes the vanishing gradient issue, allowing deep models to be trained effectively. The network learns both low level and high level features, and finally classifies the binary (normal vs diseased) with better accuracy, generalization and clinical usability.

A. Pre-processing (Low-Level Processing)

Pre-processing plays a crucial role in improving image quality and highlighting discriminative features before feeding data into the neural network.

Sobel Filter

The Sobel filter is a first-order derivative operator used for

edge detection by computing intensity gradients in horizontal and vertical directions Fig. 3. Mathematically, it applies convolution kernels:

Horizontal Kernel:

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}$$

Vertical Kernel:

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}$$

Gradient Magnitude:

$$G = \sqrt{G_x^2 + G_y^2}$$

Purpose in Retinal Images:

- Enhances blood vessels and lesion boundaries
- Extracts structural information

Advantage:

- Simple and computationally efficient
- Highlights important edge features [7], [8].

8-Neighbourhood Laplacian

The Laplacian operator is a second-order derivative filter used to detect regions of rapid intensity change Fig. ???. The 8-neighbourhood version considers all surrounding pixels. [9].

Laplacian Kernel (8-neighbourhood):

$$\nabla^2 f = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Purpose:

- Detects fine details such as micro-lesions
- Enhances high-frequency components

Advantage:

- Captures subtle abnormalities
- More sensitive than first-order filters

Log Transformation

Log transformation is a non-linear intensity transformation defined as: [11].

$$s = c \cdot \log(1 + r)$$

where c is a constant and r is the input pixel intensity.

Purpose:

- Expands dark pixel values

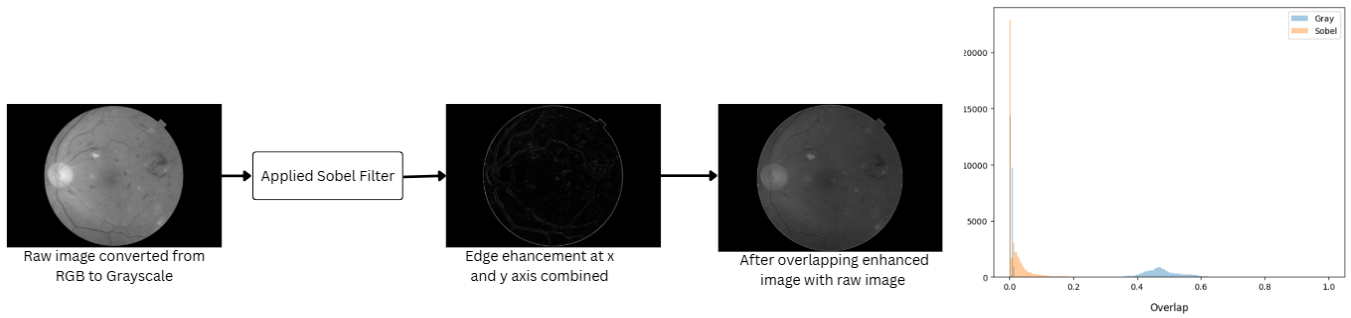


Fig. 2. Sobel Filter Application and Histogram

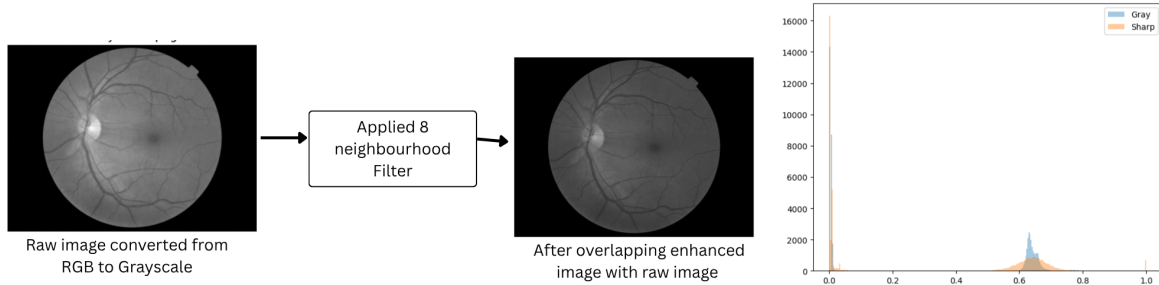


Fig. 3. 8 neighbourhood Laplacian Filter Application and Histogram

- Compresses bright regions

Advantage:

- Improves visibility of low-intensity features
- Enhances contrast in retinal images Fig. 3

B. High-Level Processing

ResNet50 is based on the concepts of a Convolutional Neural Network (CNN) and a Neural Network.

Convolutional Neural Network (CNN):

A CNN is a deep learning architecture designed for image processing tasks. It consists of:

- **Convolutional layers** → Feature extraction
- **Pooling layers** → Dimensionality reduction
- **Fully connected layers** → Classification

Neural Network:

A Neural Network is a computational model inspired by the human brain, consisting of interconnected nodes (neurons) that learn patterns from data. [10].

ResNet50 ResNet50 is a 50-layer deep residual network, which adds skip (residual) connections to overcome the vanishing gradient problem. Fig. 6 Instead of learning direct mapping, it learns: [13].

$$F(x) = H(x) - x$$

Purpose in This Work

- Conducts a binary classification (disease vs normal).
- Descends hierarchical features of already processed pictures.

Advantages of ResNet50

- Avoids vanishing gradient problem.

- Facilitates very deep architecture.
- Enhances accuracy and convergence
- Efficient feature learning

The fundamental concept of residual learning is expressed as:

$$y = F(x) + x$$

where $F(x)$ represents the residual function and x is the input. This structure allows gradients to flow directly through the network, ensuring stable and efficient training even in very deep architectures.

C. Training Strategy

The intended training plan will be used to test the effect of classical image preprocessing methods on the performance of deep learning as implemented by ResNet50. First, the model is trained on the raw RFMiD data without any preprocessing to determine a baseline and determine the ability of the network to learn discriminative features with unprocessed retinal fundus images. Nonetheless, the raw medical images are typically characterized by low contrast, noise and uneven illumination that can greatly hinder the extraction of features and overall classification.

In order to overcome these limitations, various preprocessing methods are used such as the Sobel filter, 8-neighbourhood Laplacian filter and logarithmic (log) transformation with $c = 150$ and $c = 50$. The Sobel filter boosts the edge gradients making vascular structures and lesion boundaries more visible [1], whereas the Laplacian filter highlights small details and high-frequency elements that are needed to identify micro-level abnormalities [2]. Log transformation improves darker

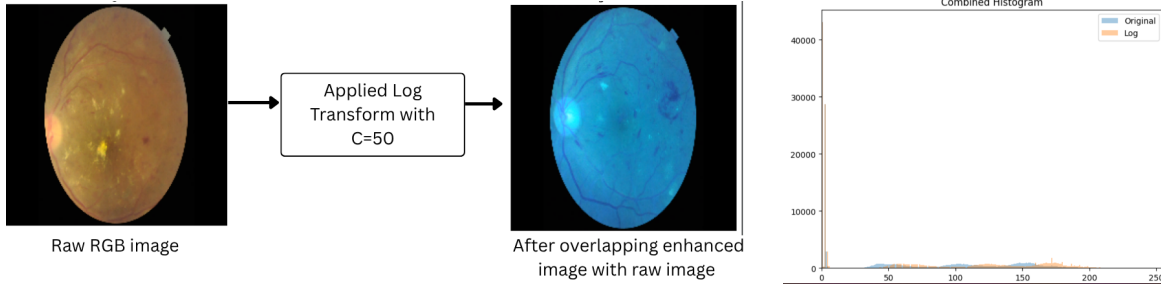


Fig. 4. Log Transform with $c=50$ and Histogram

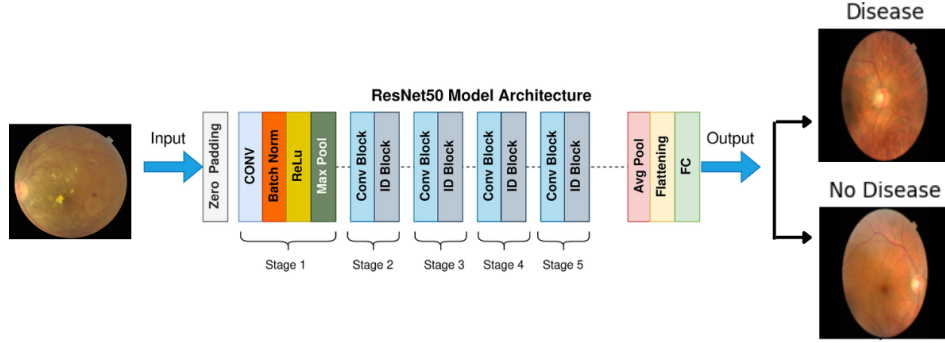


Fig. 5. Resnet 50 Architecture [12]

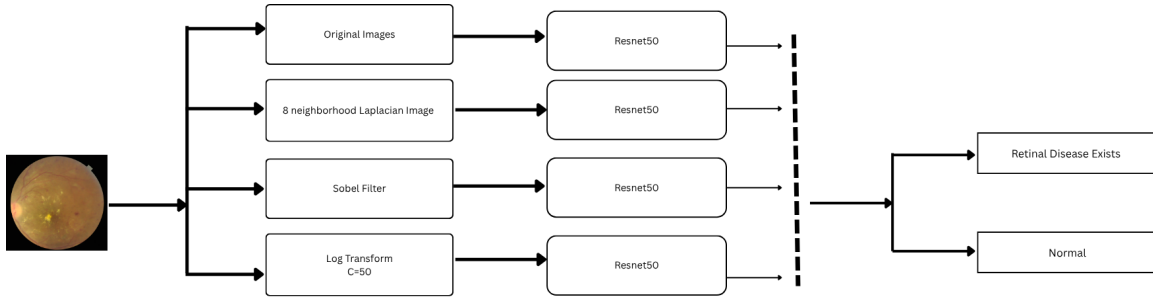


Fig. 6. Summarized Version of our Training Strategy

areas and squares brighter intensities thus improving contrast and numbers of small retinal details that might not be well seen in raw images [3]. Improved images are subsequently retrained on the ResNet50 model that makes use of residual learning to learn hierarchical features [4]. This hybrid solution will help to achieve better classification, less misclassification and a better generalization.

Moreover, RFMiD dataset is automatically multi-class and multi-label, but the work simplifies the task to binary classification. This is done using the Disease-Risk column, in which images are categorized in two sets; 0 (no disease) and 1 (presence of disease). Image IDs are matched to their respective binary labels, and there is an input-output match. This change makes the model less computationally complex, more stable in training and allows the model to specialize in distinguishing between healthy and diseased retinal conditions instead of diagnosing a specific disease as shown on Fig. ??.

D. Evaluation Metrics

The performance of the classification model is evaluated using standard metrics derived from the confusion matrix. [16] A **confusion matrix** is a utility which is used to assess the performance of classification in terms of predicted and actual values. It is comprised of four elements namely True Positive, True Negative, False Positive and False Negative. It assists in detecting various classes of errors and offers greater insight than merely accuracy particularly in medical diagnosis so as to reduce the critical misclassifications. [14], [15]

- **Accuracy:** Measures the overall correctness of the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

- **Precision:** Indicates the proportion of correctly predicted

positive cases among all predicted positives.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

- **Recall (Sensitivity):** Measures the ability of the model to correctly identify actual positive cases.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

- **Specificity:** Evaluates the model's ability to correctly identify negative cases.

$$Specificity = \frac{TN}{TN + FP} \quad (4)$$

- **F1-Score:** Represents the harmonic mean of Precision and Recall, balancing both metrics.

$$F1-Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (5)$$

where TP = True Positive, TN = True Negative, FP = False Positive, and FN = False Negative.

V. RESULTS AND FINDINGS

A. ResNet-50 without pre-processing

Without Low Level Processing : The confusion matrix of the ResNet50 model trained on the entire dataset, without any low-level preprocessing demonstrates that the model identifies all 506 cases of disease (TP) but not any cases of non-disease (TN = 0), and This is reflected in the evaluation metrics: the model has a perfect recall (1.0000), that is, it identifies all cases of the disease, but the specificity is 0.0000, which is that it is totally unable to identify healthy samples. The precision (0.7906) and accuracy (0.7906) seem to be quite large, however, they are not accurate because the model is projecting all the samples as diseased. The perfect recall also results in the F1 score (0.8831) being high even though there are no true negative predictions.

This finding warrants the fact that the model is wholly skewed towards the positive group, which is effectively a trivial classifier, and all data are classified as disease. This is possibly because of the imbalance between classes in the set or because the model has failed to pick up discriminative features of the non-disease group when only raw images are used to train the model. The model can degenerate to the prediction of the most prevalent or more learnable class to reduce loss without improvement or emphasis on features. To overcome this challenge, class weighting method, balanced sampling or threshold adjustment method should be used to punish the over-prediction of the positive class. Furthermore, appropriate preprocessing or feature improvement techniques, as well as, fine-tuning the deeper layers of ResNet50, can be used to facilitate the model to learn more balanced and discriminative representations of both classes.

The model demonstrates extreme overfitting or bias towards the positive class since it classifies all of the samples as diseased and actualizes all the recalls (1.0000) but zero specificity (0.0000). The accuracy (0.7906) and F1 score

(0.8831) are probably high but misleading since the model fails miserably when it comes to detection of non-disease cases. This is probably because of the class imbalance, and the lack of preprocessing, which does not allow the model to learn discriminative features between the two classes, instead motivating it to use a trivial prediction strategy to minimise loss. To deal with this, weighting or resampling class balancing techniques must be used and adding suitable preprocessing or feature enhancement may aid the model to learn more meaningful representations and balance performance.

B. ResNet-50 With Sobel Filter

Sobel Filter : The results of the ResNet50 mode that is trained on the entire dataset using Sobel-filtered images as shown on Fig. 8 indicate a very skewed classification performance. The confusion matrix indicates that the model makes many false negatives (245) than false positives (9) indicating that the model is often not able to identify diseased cases but it is highly conservative in predicting disease. It is also shown in the metrics of evaluation: the model has a very high accuracy (0.9667) and specificity (0.9328), i.e. when the model predicts a disease the prediction is almost always correct and the false prediction is not made very often, but the recall (0.5158) is very low. This implies that it is missing close to half of the true diseased cases, which is a severe limitation in medical diagnosis tasks where it is more significant to find the positive cases, as opposed to avoiding false alarms. The fact that the F1 score (0.6727) and the overall accuracy (0.6031) are relatively moderate, and that the model is biased towards the negative class also confirms that the model is biased.

This act could be ascribed to various factors. First, edge-enhancing filters such as the Sobel filter can suppress valuable texture and color-based information which would be important in the detection of subtle retinal abnormalities, hence resulting in lower sensitivity. Second, the model seems to prefer the majority or more easily distinguishable class which might be because of the imbalance among classes or because of the nature of features that are learned by edge-enhanced images. Also, ResNet50, with frozen layers or post-training fine-tuning, might not learn the domain-specific attributes of medical images, particularly when preprocessing distorts the underlying feature distribution.

In order to overcome these problems some improvements can be taken into consideration. To make the model more sensitive to diseased cases, it is necessary to change the decision threshold (e.g. make it smaller than 0.5), penalize false negatives more (class weighting), or it can also be done with loss functions like focal loss. Moreover, rather than using Sobel filtering exclusively, raw images could be used together with edge-enhanced images or less aggressive preprocessing techniques might be used to preserve more diagnostic details. Further layers of ResNet50 can also be fine-tuned instead of being frozen which can enhance feature adaptation. On the whole, the model is highly accurate and specific, but its low recall reveals a major drawback of this model in

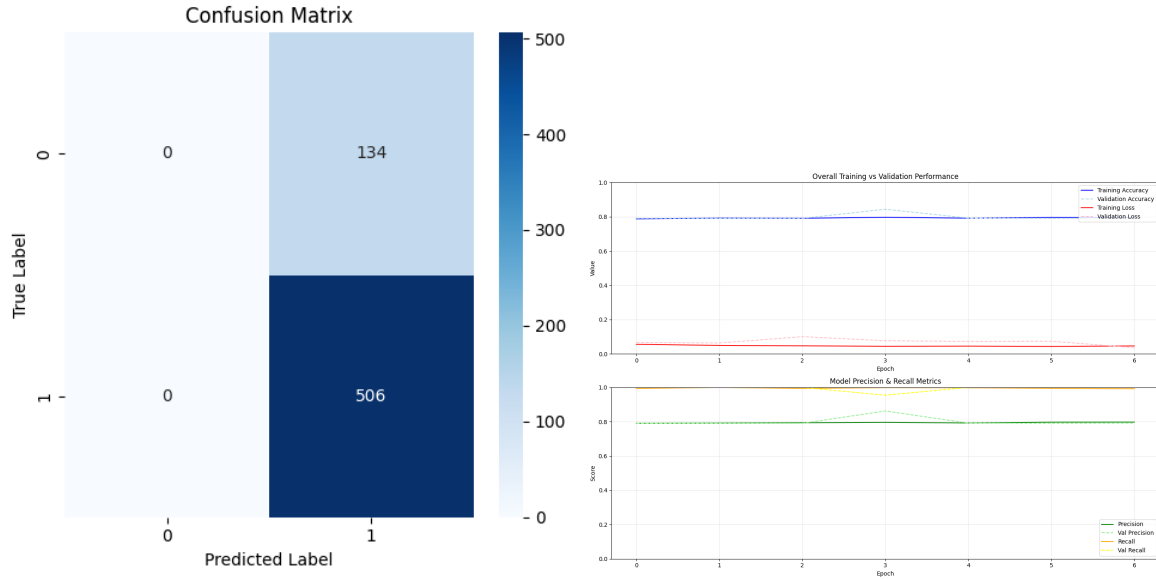


Fig. 7. Confusion Matrix and Evaluation of ResNet-50

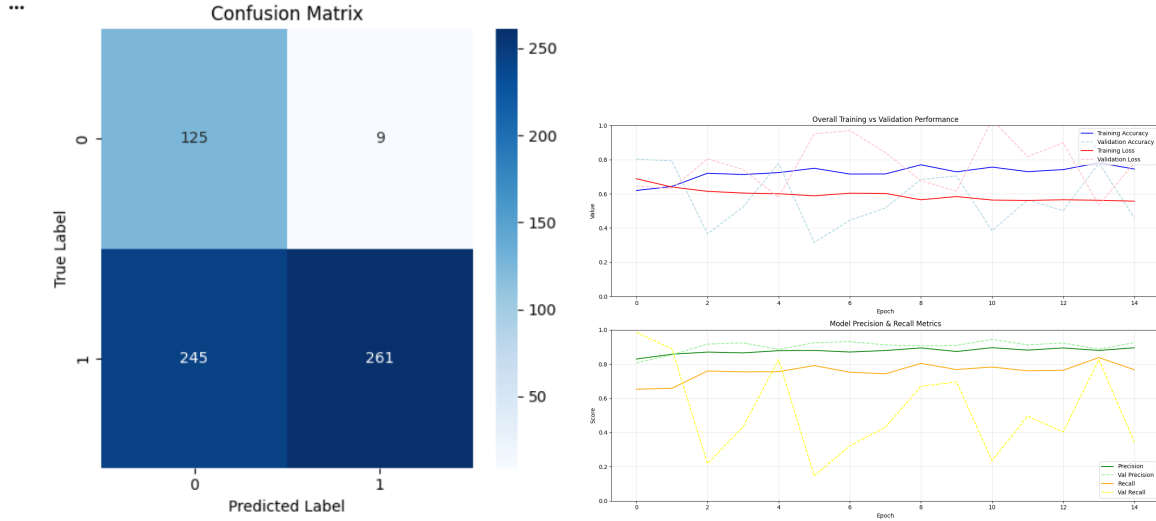


Fig. 8. Confusion Matrix and Evaluation of Sobel Filter

terms of disease detection, which may require changes and focus more on the sensitivity in this medical classification task.

This model shows that it underfits with a distinct bias towards the negative class as shown by low recall (0.5158) and high false negative (245) although the precision is high (0.9667). It implies that the model is sure of its prediction ability of disease, but it does not reflect a lot of the real disease cases. This is because the Sobel filter mainly amplifies edge information and inhibits the valuable texture and intensity features which are important in identification of small retinal abnormalities. Also, the fact that the ResNet architecture can be easily adapted to the needs of various domains can be the cause of such behavior. The finer-tuning

of deeper layers of the model, decreasing reliance on Sobel filtering or using it together with raw images and sensitivity through class weighting or threshold adjustment can be used to improve performance.

C. ResNet-50 With Laplacian Filter

The confusion matrix Fig. 9 of the ResNet50 model using the full dataset and 8-neighbourhood Laplacian filtering indicates that the model is precise in identifying 62 non-disease cases (TN) and 485 disease cases (TP) with a relatively low false negatives (21) and a high false positives (72). This finding can be seen in the evaluation measure: the model has high recall (0.9585), which means that it is able to identify nearly all cases of the disease, and a high F1 measure (0.9125) and high precision (0.8707). The specificity is low

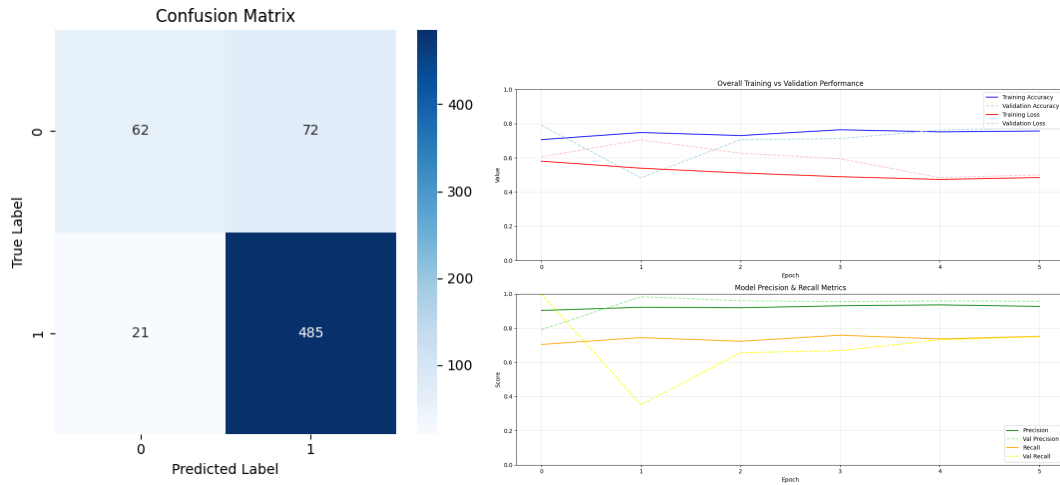


Fig. 9. Confusion Matrix and Evaluation of 8 neighbourhood Laplacian Filter

(0.4627), however, with an overall high accuracy (0.8547), the model is unable to properly classify healthy samples, and often classifies them as diseased. This finding supports the fact that the model is very sensitive to the detection of disease but has a biasness towards the over-prediction of the positive class. This is because the 8-neighbourhood Laplacian filter boosts edge and high-frequency detail, which increases the visibility of the pattern of disease-related patterns and increases their ability to be detected, hence the high recall. Nonetheless, it enhances noise and normal retinal images, which makes the model to confuse normal images with diseased images, resulting in high false positive values and low specificity. To overcome this problem, one can enhance the model by changing the classification threshold to minimize false positives, use class balancing methods, or a combination of Laplacian-filtered images and original images to keep more contextual information. This can also be done by fine-tuning more layers of the model and ensuring that the model can better differentiate between the real pathological features and noise.

This model has a slight overfitting to the positive class, very high recall (0.9585) but low specificity (0.4627), which means that the number of false positives (72) is large. Although the model is effective in detecting most of the cases of diseases, it fails to identify healthy samples correctly. This is because the Laplacian filter boosts the high-frequency content and as such the disease-related features are more pronounced, however, noise and normal retinal structures are also boosted and this results in erroneous classification. To deal with this, the decision threshold of the model can be raised to decrease false positives, regularization methods like dropout can be reinforced, and the use of filtered images alongside original images can assist in maintaining contextual information and improving the balance in classes.

D. ResNet-50 with Log Transform for variable values of c ($c=50$)

Log Transform $C=50$ The confusion matrix of the ResNet50 model that was trained on the full dataset with log transform ($c = 50$) indicates that the model is accurate in identifying 134 non-disease cases (TN) but not a single disease case (TP) and the number of false negatives (505) is enormous, and the number of false positives is zero. From Fig. 10 is evident in the assessment scores: the model has a perfect precision (1.0000) and specificity (1.0000), i.e. it does not predict disease in healthy samples, the recall is very low (0.0020), i.e. it does not accurately predict a disease case. This gives a very low F1 score (0.0039) and general accuracy (0.2109) indicating that the model is highly skewed to predicting the negative class. This finding explains why the model has essentially learned to categorize almost all inputs in the non-disease category, and has not found significant patterns in the positive category. This is probably due to the fact that the log transformation with a high constant ($c = 50$) changes the intensity distribution of the images dramatically, compacting the significant changes in features in the images and rendering the patterns associated to the diseases less pronounced. Consequently, the model ends up being too conservative and is not able to predict the positive class at all. To overcome this problem, a smaller value of c must be applied in order to keep more valuable image information or other preprocessing procedures should also be thought over. Moreover, sensitivity and better disease case detection can be attained by further refining the model using the classification threshold, class-weighting, or by simply fine-tuning the model.

The model has an extreme underfitting with a very low recall (0.0020) and a very large number of false negatives (505), although perfection in the model (1.0000) and specificity (1.0000). This means that this model nearly always classifies under the negative category, and does not learn any meaningful features that are disease-related. This is due to the aggressive

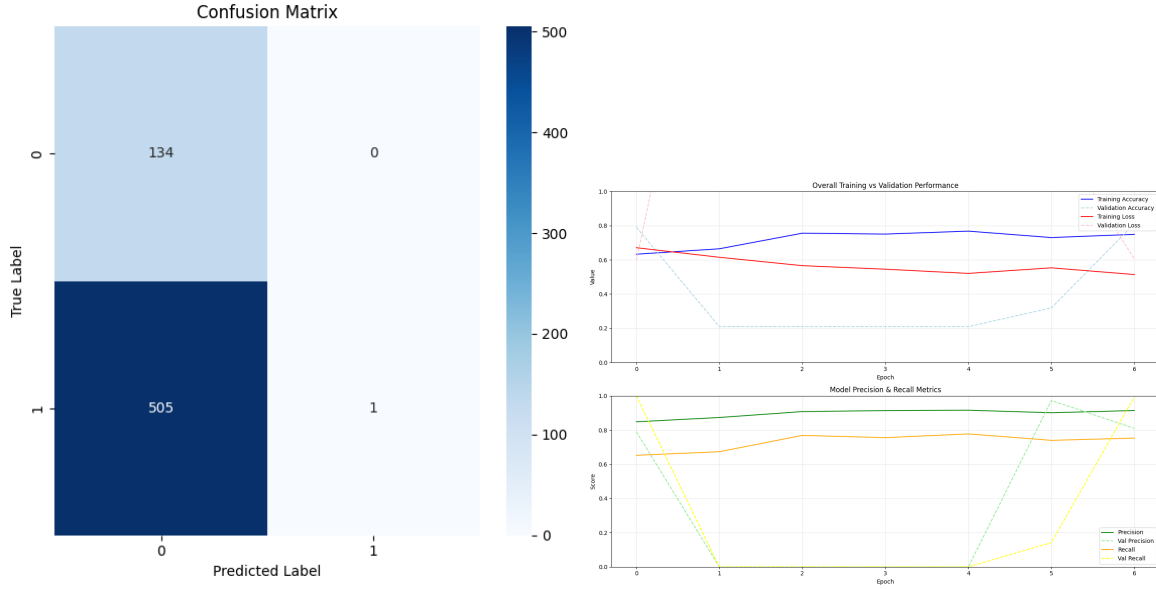


Fig. 10. Confusion Matrix and Evaluation of Log Transform C=50

log transformation having a large constant value that causes compression of the image intensity range and elimination of important variations that are required to extract features. Consequently, the model is not able to classify classes adequately. The transformation constant value should be decreased or the preprocessing step rethought and the model fine-tuned to better fit the data.

A comparison of all model are given on Table below :

TABLE III
MODEL PERFORMANCE COMPARISON (RESNET WITH DIFFERENT PREPROCESSING)

Metric	Sobel	8 Neighbourhood Laplacian	Log Transform (c=50)	No Low Level Processing
Accuracy	0.603	0.855	0.211	0.791
Precision	0.967	0.871	1.000	0.791
Recall	0.516	0.959	0.002	1.000
Specificity	0.933	0.463	1.000	0.000
F1 Score	0.673	0.913	0.004	0.883
Issue Type	Underfitting	Slight Overfitting	Severe Underfitting	Overfitting (Bias)
Key Problem	Misses disease	False positives	No detection	Predicts all disease

VI. CONCLUSION

This study highlights how important image preprocessing is in improving the performance of deep learning models for retinal disease classification. The ResNet50 model yielded medium-level results when applied to raw images, primarily because of such problems as noise, low contrast, and varying quality of images, which are typical in medical imaging. Using classical preprocessing methods, the image of the important features in the retina became much more visible, which led to a further increase in the overall performance of the model.

Each approach had its advantages and disadvantages. The Sobel filter offered consistent edge detection and less noise which was beneficial to the model generalizing, but failed to capture very fine details. Laplacian filter was also more sensitive to minor features such as micro-lesions, but this filter also increased the noise and had to be used carefully.

The log transformation enhanced contrast particularly in dark areas bringing latent structures to fore. The above trade-offs demonstrate that none of the approaches is perfect alone.

In general, preprocessing, combined with a deep learning framework, such as ResNet50, results in improved accuracy, detection of diseased cases, and reliable performance. This paper has shown that the combination of an integrated approach is central to the development of effective, scalable, and clinically helpful retinal disease detection systems.

VII. CONTRIBUTION

This paper is a comparative study of preprocessing methods combined with a deep learning system to classify retinal images. Raisa Tafannum used 8-neighbourhood Laplacian operator with ResNet-50 to improve the extraction of features using edges and better representation of structural details. Jugumaya Saha Disha created a similar scheme with the Sobel filter and ResNet-50, aiming at gradient-based improvement of the features to detect the boundary better. Besides, the authors also jointly transformed the images with logarithmic transformation ($c = 50$) to normalize their intensity and improve the low-contrast areas and then classify them with ResNet-50. A performance baseline model was also created with ResNet-50 without any preprocessing. The contributions together allow a systematic analysis of the effects of various spatial filtering and intensity transformation methods on the performance of deep models, providing information on the best preprocessing methods to achieve high accuracy and robustness in retinal disease classification tasks.

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